

ARNAV KUMAR SHUKLA

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EDUCATION

Indian Institute of Technology Bombay
Master of Science – Statistics — CPI: 8.68 / 10

Mumbai, India
2025 – Present

Banaras Hindu University (BHU)
Bachelor of Science – Statistics — CGPA: 8.13 / 10

Varanasi, India
2022 – 2025

WORK EXPERIENCE

AI Engineer Intern | *Smart Analytica*

(May'25 – Jul'25)

- Built a **Test Shortlister** that embeds issue descriptions via **nomic-embed-text-v1.5** and scores **60+ diagnostic tests** using **cosine similarity** against a 300+ record pgSQL description table, with engine-specific fallback defaults
- Authored **60+ standalone diagnostic tests** spanning **Hive, Impala, and Spark**, covering failure and slowness signatures (skew, spill, small files, resource starvation, stuck stages and GC pauses) under a consistent I/O contract
- Co-designed a two-track **RCAOrchestrator** pipeline (Exception vs. Slowness) with diagnostic testings triggering selective expansion rounds when fewer than 2 failures found, feeding KB matching & **LLM-based** root cause analysis
- Contributed across the pipeline from data collection through report assembly, including **KB matching, custom LLM prompt** integration, & the persistence layer, with automatic fallback to raw KB matches on LLM call failure

KEY PROJECTS

RAG-QA Hallucination Detection System | *Self Project*

 *(Feb'25 – May'25)*

- Built a **RAG pipeline** over PDFs using **ChromaDB, Sentence Transformers, & Cohere Rerank** for retrieval
- Generated with **GPT-4o-mini** via LangChain & detected hallucinations using a **DeBERTa NLI cross-encoder**
- Implemented confidence-based verdicts with sentence-level contradiction detection and evidence scoring for reliability
- Deployed the system as a **FastAPI** REST service with secure credential management and Swagger documentation

Agentic Stock Market Research Assistant | *Self Project*

 *(Mar'25 – May'25)*

- Designed a **LangGraph** multi-agent pipeline with 6 specialised nodes: ticker resolution, market data via **yFinance** news via **Tavily**, sentiment analysis, signal generation, and report writing for stock analysis and investment decisions
- Implemented **adaptive retry logic** that re-queries news on low-confidence sentiment before recommendation engine
- Generated BUY/HOLD/SELL signals using **GPT-4o-mini** with **Pydantic v2** built a **Streamlit** frontend reports

Encoder-Decoder NMT with Bahdanau Attention | *Self Project*

 *(Nov'24 – Jan'25)*

- Implemented a **Seq2Seq Encoder-Decoder** in **TensorFlow** for English→French translation with LSTM encoder decoder and custom **Bahdanau Attention** using teacher forcing during training with attention visualization support
- Applied **Teacher Forcing** on **100K+** parallel sentence pairs; built separate inference models with greedy decoding
- Trained two variants (with and without attention) and validated translation quality on held-out pairs using inference

Fraudulent Transaction Detection | *Self Project*

 *(Sept'24 – Dec'24)*

- Preprocessed **284K+** credit card transactions via **PCA** and **StandardScaler** with stratified 80/20 train-test data split
- Compared **SMOTE & RandomUnderSampler**; tuned thresholds to balance precision-recall tradeoff across classes
- Trained **Random Forest** and **XGBoost**, achieving **Recall of 0.84, F1-score of 0.83**, and **ROC-AUC of 0.96** applied **K-Means** for unsupervised fraud pattern discovery with **Silhouette Score of 0.83** after SMOTE balancing

Named Entity Recognition & POS Tagging using BiLSTM | *Self Project*

 *(Jul'24 – Sept'24)*

- Trained separate **BiLSTM** models in **TensorFlow/Keras** for POS tagging and NER on the **GMB** corpus with a custom **masked accuracy** metric for padded sequences using trainable word embeddings and masking of sequence
- Achieved **96.4%** masked accuracy on POS tagging and **95.4%** on NER; architecture: Embedding → BiLSTM → TimeDistributed Dense → softmax with token-level sequence classification outputs across variable-length sentences
- Built a unified inference pipeline for parallel POS and NER prediction on raw input sentences with batch inference

TECHNICAL SKILLS

Languages: Python | R | SQL **Interests:** Machine Learning | Generative AI | Agentic AI | NLP | Deep Learning
ML / DL: PyTorch | TensorFlow/Keras | Scikit-Learn | XGBoost | HuggingFace Transformers
LLM / Agentic: LangChain | LangGraph | OpenAI API | Cohere API | FastAPI | ChromaDB
Big Data: Apache Spark | Hadoop | Hive | YARN **Tools:** Git | Google Colab | VS Code | $\LaTeX$

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 33** among **2880** students in **IIT JAM Mathematical Statistics Examination (2025)**

POSITION OF RESPONSIBILITY

Interview Coordinator | *Institute Placement Team, IIT Bombay*

(Oct'25 – Dec'25)

- Facilitated placement of **2300+** students collaborating with a team of **250+** members across all departments
- Coordinated assessment tests for leading firms & managed end-to-end placement logistics efficiently throughout drives